INCH POUND

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COMMERCIAL ITEM DESCRIPTION

TRACTOR, AIRCRAFT TOWING, FLIGHT LINE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers the requirements for one type of Flight Line Tow Tractor (FLTT), hereafter referred to as tow tractor. The tow tractor is based on a commercial truck chassis, also known as a chassis cab, which is modified for use as a tow tractor in worldwide service. These chassis are generally sold to final-stage manufacturers (hereafter referred to as tow tractor manufacturers) who complete the vehicle. The tow tractor is capable of towing trailers up to 40,000 pounds, and military aircraft up to 70,000 pounds on grades up to two and one half percent.

2. SALIENT CHARACTERISTICS. Failure of the tow tractor to meet the salient characteristics of this CID shall be cause for rejection of the tow tractor. The tow tractor shall have a life expectancy of 15 years, and shall be free from faulty construction and any defect that affects appearance, serviceability, and durability. The tow tractor shall be representative of the quality and workmanship found in tow tractors offered for new commercial sale.

2.1 Gross vehicle weight (GVW). The tow tractor shall have a minimum GVW of 8000 pounds. However, the GVW shall not exceed gross vehicle weight rating (GVWR) less l500 pounds. The minimum GVWR required for this application is 9500 pounds.

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2.2 Overall dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed:

Length: 200” without couplers
Width: 100”
Height: 100”

2.3 Drawbar pull (DBP). The tow tractor shall have a minimum DBP of 4,500 pounds on dry level concrete.

2.4 Power train. The tow tractor shall have a diesel engine of not less than six cylinders, automatic transmission, and power train sufficient to develop the required drawbar pull and obtain the required tow tractor speeds. Power train components shall have adequate torque and power ratings for the imposed torque and power. The tow tractor power train shall possess sufficient cooling capacity to perform continuous maximum performance towing in high ambient temperatures of +125 degrees F without exceeding the Original Equipment (OE) Chassis-Cab Manufacturer’s (M) maximum allowable temperatures on the individual power train components. Any requirements for towing in high ambient temperatures (recommended coolant ratios, oil types and viscosities, etc., for various temperature ranges) shall appear in the Technical Order (TO). The diesel engine shall operate satisfactorily when using Jet A-1 (JP-8) fuel. Power train shall feature a standard (type C) winterization package that is good for starting down to minus 20F, or when specified (see 6.2 c), a heavy-duty (type A) winterization package good for storage down to minus 65F and starting to minus 40F. Winterization package components with external power requirements shall be compatible with 110VAC/20A circuitry. For each winterization package, a breakdown of the winterization components and requirements (recommended coolant ratios, oil types and viscosities, etc., for the various temperature ranges) shall appear in the TO. The tow tractor shall meet applicable 50-state exhaust and evaporative emissions standards in effect at the time of vehicle manufacture. The tow tractor shall include an anti-restart, as well as a neutral safety start feature. The rear axle shall have a limited-slip differential and a reduction ratio equal to or greater than 4.1:1. The rear axle shall feature single rear wheels. A “special option” from the OE chassis-cab manufacturer may be required to meet this axle ratio specification.

2.5 Driveline and steering. When specified (see 6.2 f), the tow tractor shall be offered in a part-time, four-wheel drive variant. The rear axle reduction ratio shall be the same on the two-wheel drive and the four-wheel drive tow tractor variants. Non-directional mud and snow tires are required on the four-wheel drive variant. The tow tractor shall have a maximum wall-to-wall turning radius of 21 feet or less. The wheelbase shall be between 90 and 100 inches. The balance of these two parameters shall be optimized.

2.6 Electro-magnetic interference (EMI). The tow tractor shall comply with EMI requirements of MIL-STD-461 for Radiated Emission per RE102, Radiated Susceptibility per RS103, and Conducted Susceptibility per CS114, CS115, and CS116 for Air Force ground equipment.

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2.7 Mobility. The tow tractor shall be capable of 35 miles per hour (MPH) (minimum) on flat asphalt or concrete surfaces without a trailing load. The tow tractor shall be governed not to exceed 40 MPH (plus 0, minus 3), unless the rear axle reduction ratio prevents speeds greater than 40 MPH. The preferred method to control the tow tractor speed is through the selection of the axle gear ratio. Reverse speed shall be a minimum of 2.5 MPH on flat surfaces. The tow tractor shall be capable of towing trailers from a stop to 10 MPH (minimum) and aircraft from a stop to 5 MPH (minimum) on grades up to 2.5% on asphalt or concrete surfaces. The tow tractor shall be capable of operating on 10 percent side slopes and 12 percent grades without fluid spillage.

2.8 Suspension. Reserved.

2.9 Wheels and tires. The tow tractor shall be equipped with steel disc wheels and tubeless steel belted radial tires with an appropriate tread pattern. The valve stems may be rubber or metal, but the valve stem caps shall be vinyl or plastic. The wheels and tires shall be in accordance with (IAW) the Tire and Rim Association recommendations for this application. All wheels and tires shall be uniform. Tire inflation pressures shall be noted on all wheel well openings, and shall be in accordance with Air Force Technical Order (T.O.) 36-1-191. If tire loads and tire inflation pressures that exceed the specifications on the tire sidewall markings are required, a copy of the application approval from the tire manufacturer shall be added in the Technical Order.

2.10 Braking system. The tow tractor shall feature a braking system in compliance with Federal Motor Vehicle Safety Standards (FMVSS) for vehicles of this GVWR.

2.11 Parking brake. The tow tractor shall be equipped with a parking brake that shall be in compliance with SAE J293. The tow tractor manufacturer shall ensure that any parking-brake cable relocation or cable shortening (resulting from modifying/shortening the wheelbase of the commercial truck chassis) is equivalent to the OE chassis-cab manufacturer’s design in providing parking brake cable tension.

2.12 Air system. When specified (see 6.2 d), the tow tractor shall have an air system for towing trailers with air brakes. The air system shall be an over-the-road commercial truck/tractor design to include a reservoir tank with heated and automatic drain valves, automotive compressor, dryer, trailer protection valve, low-pressure indicator, service and emergency glad hands, glad hand covers with lanyards, etc. The glad hands shall be in accordance with SAE J318, and shall be polarized and identified as “service” and “emergency”. Components of the air system that are sensitive to the elements (rain water, road water, ice build-up, road grit, etc.) and installed in positions on the tow tractor that are vulnerable to those elements shall be covered or shielded to prevent the premature failure of those components over time or during service. The configuration of the air system shall be the same on the two-wheel drive and the four-wheel drive tow tractor variants.

2.13 Electrical system. The tow tractor shall be equipped with a 12V electrical system. The battery(ies) shall be maintenance free. All external chassis wiring shall incorporate weather-

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proof connections and terminations. Exceptions to this are under-hood terminal strips and electrical posts that are protected from the elements. Any wire splicing (typically resulting from special-request options) shall incorporate a mechanically-crimped connector, solder and heat-shrink tubing. All crimped-on wire components (i.e. Weatherpac or Metri-pac style connectors) shall be installed using the appropriate crimping tool for the application. All wiring shall be of the appropriate gauge for the amperage the circuit is drawing and/or protected to. Any holes added to the body or chassis for wiring to pass through shall incorporate a grommet.

2.14 Chassis modifications. The commercial truck chassis shall be modified to the specified wheelbase by relocating the rear axle, suspension, and other affected components forward on the original chassis frame rails. Back-halved chassis or spliced-in rear sub-frames, creating a joint in the center of the chassis length, shall not be used in the modification. Chassis components shall be relocated and/or modified as necessary to avoid excessive loading, suboptimal drive angles, and driveline vibrations. Modifications to chassis components, especially load bearing suspension components, shall be performed using an engineering approach and with quality workmanship so as to avoid weakening the structural integrity of the component. Excess length at the rear of the chassis frame rails from the forward relocation of chassis components shall be removed using a process that leaves a smooth cut finish. Raw metal exposed by the modifications shall be primed and painted chassis color. The relocation of chassis components shall not result in improperly loaded fasteners. Modifications to the truck chassis shall not result in exposed fuel supply or fill lines. During the wheelbase-modification process, if the tow tractor manufacturer determines that flexible fuel-line splicing is necessary into the OEM metal fuel lines (such as when routing flexible fuel lines to a new fuel tank location), then at a minimum, fuel injection-rated hoses must be utilized, with the ends of the metal lines modified with barbed or flared ends to prevent the flexible fuel lines from blowing off of the metal fuel lines under pressure. Fuel injection-rated hose clamps must also be used. Alternatively, OEM-quality, flexible fuel-line connections can be accomplished with the ends of the OEM metal lines cut and flared after a tube nut has been slid over the end of the cut metal line. The flared metal line and integral tube nut can then be threaded into a female-inverted, “field-attachable” brass fitting and tightened. These fittings are available from companies such as Eaton-Weatherhead as a 100 ‘B’ Series Barb-Tite fitting, Parker-Hannifin as a Push-Lok 82 Series fitting, and Aeroquip as a Socketless fitting. These type applications do note require a hose clamp when the appropriate working pressures are observed, and the correct flexible hoses for the application are selected. Any holes added to the chassis for fuel lines to pass through shall incorporate a grommet. The exhaust pipe shall exit on the right-hand side, over the rear axle and behind the rear wheel, oriented at 90 degrees from the centerline of the vehicle. If any information found on the factory-original commercial truck chassis data plates (normally found in the door jambs) is not consistent with the modified commercial truck chassis design, remove and replace them with data plates that contain updated and correct applicable information (such as rim and tire sizes, tire air pressures, GVW, etc.) that pertains to the new design.

2.15 Cab design. The tow tractor shall be equipped with a three-man cab. The first entrance step to the cab shall be a full-length, running board style step and shall not exceed 18 inches in height from the ground. The outer edge of this running board style step shall be offset out from the outer edge of the cab door opening by at least 4.75 inches. Cab instrumentation shall feature an hour meter in addition to the standard instrumentation. When an air system for towing trailers

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is specified (see 6.2 d), cab instrumentation shall include an air pressure gauge and a low-pressure indicator with buzzer and/or warning light. The tow tractor shall have a covered toolbox with a hinged lid and latching mechanism located behind the cab in an easily accessible location. The toolbox lid shall be supported when opened, and the latching mechanism shall not bind when the toolbox lid is closed. When specified (see 6.2 h), the tow tractor shall have an Air Conditioning system. The tow tractor cab shall be equipped with wipers, a windshield washer, driver and passenger-door rear-view mirrors, windshield-area rear-view mirror, visors, heater/defroster, interior dome light, seat belts, and other equipment as found in standard commercial offerings. The cab shall feature a rear sliding window and a 12V power outlet (cigarette lighter type).

2.16 Lighting. The tow tractor lighting shall be equivalent to the standard commercial offering to include a rear adjustable aim floodlight, parking lights, head lights (high and low beam), brake lights, turn signals, hazard lights, and reverse lights. Basic lighting system shall be in compliance with FMVSS requirements. A flashing amber beacon light shall be mounted above the roofline on the tow tractor's centerline. Two double-faced signal lamps (SAE J588) shall be mounted above the roofline, approximately 48 inches apart and centered on the tow tractor's centerline, and shall be integrated with the tow tractor's taillights. The flashing beacon light and double-faced signal lamps shall be mounted on a light bar that is not fastened to the cab that runs approximately one inch above the roofline, to avoid drilling and cutting into the cab. The rear adjustable aim
floodlight, of not less than 3000 candlepower, shall be mounted on the tow tractor so that the rear working area behind the cab and the rear coupling positions are illuminated. When specified (see 6.2 g), the tow tractor shall be equipped with an adjustable spotlight, capable of illuminating 180 degrees of elevation and 360 degrees of rotation, and located on the driver's side of the cab. The spotlight shall be installed in the driver's door-to-windshield pillar, and steerable from the driver's position, inside the cab.

2.17 Pintle hooks. The tow tractor shall be equipped with four pintle hooks that incorporate a cotter or snapper pin as a secondary-locking device, and conforming to CID A-A-52550, Type 1, Class 2 (old MS51335-2). Pintle hooks shall be attached to the pintle hook mounting structure with a minimum of four ½-13 or ½-20, Grade 8 fasteners. The pintle hook fasteners shall be installed with a design fastening torque applied such that an adequate preload is introduced in the fastener to prevent losing or fatiguing the fasteners and/or the pintle hook. A description of the fastener type and the fastening torque shall be recorded in the Technical Order.

2.17.1 Front pintle hook position. The front of the tow tractor shall be equipped with one center-mounted pintle hook capable of withstanding a minimum of 18,000 pounds DBP, with the coupling position of the pintle hook located at 18 ½ inches, plus or minus ½ inch, from the ground. Additionally, the front pintle hook assembly and mounting structure (all structural components inclusive between pintle hook and frame) shall comply with the following:

a. The rated load shall be the load combination of forces that the pintle hook assembly and structure must support or resist in a static state.
b. The dynamic load shall be the rated load with adjustments for loads and accelerations, in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate, demate, load, and unload operations.

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c. The design load shall be based on the rated load multiplied by a factor of three, or on the dynamic load multiplied by a factor of two, which ever is greater. This design load shall be considered the minimum load for attaining the design stress levels.

2.17.2 Rear-outboard pintle hook positions. The rear of the tow tractor shall be equipped with two outboard pintle hooks capable of withstanding a minimum of 6,000 pounds DBP, located in the same vertical plane, with the pintle hook centerlines at 43 ½ inches, plus or minus ½ inch, to each side of the tow tractor centerline, with the coupling positions of the pintle hooks located at 18 ½ inches, plus or minus ½ inch, from the ground. The rear-outboard pintle hook assemblies and mounting structures shall be designed to accommodate the greatest force exerted by the drawbar with adjustments for loads and accelerations, in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate, demate, load, and unload operations.

2.17.3 Rear-center pintle hook position. The rear of the tow tractor shall be equipped with one center-mounted, extendable pintle hook capable of withstanding a minimum of 6,000 pounds DBP in the extended position, a minimum of 18,000 pounds DBP in the retracted (closed) position, and with pintle hook coupling positions that are height adjustable from 12 inches to 18 ½ inches from the ground. A minimum of four evenly spaced or coupling positions (including 12 inches and 18 ½ inches) shall be provided. Height adjustment shall be easily made by the operator without the use of tools. The rear-center extendable pintle hook shall not extend more than 10 ½ inches beyond the rear plane of the truck when retracted for towing. The extendable unit shall consist of a self-locking sliding bar that shall allow the hook backward movement of not less than 10 ½ inches, or more than 12 inches, and a lateral swing movement of 45 degrees in either direction from the centerline of the tow tractor. The unit shall be fitted with rollers, and provisions shall be made for lubricating all moving parts. The normal towing position shall be with the pintle hook/hitch assembly in the retracted and straight positions, rather than the extended and swung positions. The extendable bar shall be equipped with a safety grab handle, attached to the pintle hook/hitch assembly, and positioned above the pintle hook, to allow for easy user access and function of the extendable hitch's front-to-rear and side-to-side hitch motion, as well as the pintle hook height adjustments. The extendable bar shall be easily removable to facilitate inspection and maintenance. The extendable/swing features are required to facilitate connection of the pintle hook to the trailer by a single operator, rather than a method requiring two users. Once the lunette or towing eye of the towed load is secured, the extendable unit shall be engaged in the closed and straight positions for towing by the backward movement of the tow tractor until the self-locking device is secure. Any shear pins, locking pins, or related pintle hook components that are designed to be removed for adjustments by the user shall be attached to the tow tractor with lanyards. Removable items that are heavier (such as the pintle hook assembly if applicable) shall be attached with an appropriate, heavier gauge lanyard. Those shear pins that use a locking pin for retention shall utilize a locking pin that is a double-loop hair-pin design. If constructed of mild steel, the hitch shall have hardened steel bushings and pins at all pivot and locking points. The bushings shall be easily removed for service. Additionally, the rear-center pintle hook assembly and mounting structure (all structural components inclusive between pintle hook and frame) shall comply with the following:

a. The rated load shall be the load combination of forces that the pintle hook assembly and structure must support or resist in a static state.

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b. The dynamic load shall be the rated load with adjustments for loads and accelerations, in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate, demate, load, and unload operations.
c. The design load shall be based on the rated load multiplied by a factor of three, or on the dynamic load multiplied by a factor of two, which ever is greater. This design load shall be considered the minimum load for attaining the design stress levels.

2.18 Breakaway cable hookup. The tow tractor shall be equipped with a rear trailer emergency breakaway cable hookup, capable of mating with NAS 1281C10 spring snap hook. The breakaway chain attachment shall be installed on the bumper or frame, on the tow tractor's centerline, within 8 inches (vertical distance) above the top of the center pintle hook while in its uppermost height position, and within 4 inches forward of the base of the pintle hook. It shall not be installed on the pintle hook or pintle hook assemblies.

2.19 Safety chain attachment points. The tow tractor shall have two safety chain attachment points, one located on each side of pintle hook, symmetrical about the pintle hook, and located at a distance of 26 inches in height from the ground for use during towing on the center, rear pintle hook only. The vertical centerline of the safety chain attachment points shall be located at a parallel distance of not greater than 6 inches to the left and right of the tow tractor's centerline, and shall in no way be a part of or attached to the pintle hook. Additionally, the rear edge of the safety chain attachment points shall be within 2 inches forward of the base of the pintle hook. Each of the attachment points shall be capable of withstanding a maximum shock load of 5,000 pounds and shall have the ability for standard safety chain hooks to be attached. The inside radius of the safety chain hook is between one inch and three inches.

2.20 12-Volt Trailer light harness receptacle. The tow tractor shall be equipped with an SAE J560b electrical receptacle, provided on the tow tractor's centerline, within 12 inches (vertical distance) above the top of the center pintle hook while in its uppermost height position, and within 15 inches forward of the base of the pintle hook. All terminals of the receptacle shall be wired in accordance with SAE J560b. The tow tractor shall be equipped with an aftermarket, heavy-duty trailer-light circuit protection system that shall isolate the trailer's lighting circuits from the tow tractor's electrical system. This system shall power any trailer lights directly from the tow tractor's battery, and be capable of handling up to 20 amps. The #7 terminal (blue wire) in the J560b connector shall be an auxiliary, ignition-hot circuit, also capable of handling 20 amps.

2.21 Fire extinguisher. The tow tractor shall be equipped with two 5 pound (minimum) dry chemical type fire extinguishers with mounting brackets, and a ULC rating of 4A:20B:C (minimum). Extinguishers shall be mounted on the left and right side of the tow tractor, rear of the cab.

2.22 Finish. Prior to the application of the finish top coat, any body or chassis component specifically manufactured by or for the tow tractor manufacturer, from sheet, plate, angular, or tubular steel, shall be primered with a zinc-rich primer. As specified (see 6.2 e), the tow tractor shall be painted Dark Green (Color 14052, FED-STD-595) or Desert Sand (Color 23448, Fed-STD-595) using offeror's standard commercial paint system. A "special option" may be

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available from the OE chassis-cab manufacturer to have the chassis cab delivered to the tow tractor manufacturer in the specified paint colors above, eliminating the need for the tow tractor manufacturer to paint the cab, and providing an OEM-quality finish top coat. If this "special option" cannot be obtained, chassis cabs shall be ordered from the OE chassis-cab manufacturer in a color that is close to the final color that the tow tractors will be painted (i.e. ordered in black for a final finish coat in the dark green color, and gold for a final finish coat in the desert sand color). Prior to applying the finish top coat, the chassis cab's OEM finish shall be properly prepared by surface conditioning (scuffing with an abrasive material and cleaning) the paint at a minimum. The finish top coat shall extend into the door jambs and hood jambs to the extent that the OEM finish color cannot be seen through the gaps between the body panels when the doors and hood are closed.

2.23 Corrosion protection. The tow tractor shall be corrosion-protected with a thick, hard, black, commercially-available Corrosion Preventive Compound (CPC) that is difficult to remove. Application shall include all of the body-chassis metal (both OEM and added) underneath the tow tractor (exposed, boxed-in, and enclosed areas), the inside of the wheel wells, and behind the fender skirts. Tow tractors that are to be shipped overseas shall be temporarily corrosion-protected from salt spray with a thin, soft-film, semi-transparent, commercially-available CPC that is removable with a low pressure steam. Application shall consist of the exterior of the tow tractor in its entirety. The overseas Air Force vehicle maintainers shall remove this temporary CPC on receipt of the tow tractor from the shipper. The Commercial Tow Tractor Manual shall reflect the temporary CPC removal process.

2.24 Visibility. The tow tractor shall be so constructed as to provide maximum visibility in all directions for the 95th percentile driver. The three rear pintle hooks shall be visible either directly or via mirrors from the driver's position. If required, one mirror per pintle hook shall be provided to complete this visibility requirement. A plane defined by a line 18 inches above the rear pintle hooks and a point represented by the driver's eye shall not be obstructed by any structure or feature on the truck. The light bar is exempt from this requirement.

2.25 Air transportability. The tow tractor shall be air transportable on C-130, C-141, C-17 and C-5 aircraft without removing ballast or shoring to load the aircraft (referred to as approach shoring and step-up shoring). Use MIL-HDBK-1791 as a guide. In all transport configurations, the tow tractor shall be capable of being restrained and withstanding 2.0 G up, 4.5 G down, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The tow tractor shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure.

2.25.1 Air transportable configuration. The air transportable configuration shall include all equipment, ¾ tank of fuel, and no crewmembers. At least 6.0 inches sidewall and 6.0 inches overhead clearance shall be maintained between the tow tractor and the aircraft at all times during loading and flight. The restrained tow tractor shall allow for loadmaster in-flight access from the front to the rear of the aircraft.

2.25.2 Equipment removal/reconfiguration. Preparation and restoration of the tow tractor for air transportability shall take no more than 15 minutes for two persons using common non-powered

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hand tools. All equipment removed shall be stored on the tow tractor; caps and plugs shall permit driving and storage in transport configuration.

2.25.3 Tie down points. The tow tractor shall be symmetrically restrained during air transport. Tie down points shall be rated at a minimum of 10,000 pounds, marked for capacity, with a clear opening compatible with the appropriate devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the tow tractor.

2.25.4 Shoring. The use of shoring during flight (referred to as load shoring, parking shoring, and sleeper shoring) is permitted, but not desired. However, the use of shoring is not an alternative to good tow tractor design.

2.26 Transportation data plate. The tow tractor shall be provided with a permanently-marked transportation data plate constructed of a non-corroding metal, mounted on the exterior surface of the tow tractor. The plate shall contain at least the following information:

1. Side and rear silhouette views of the tow tractor
2. Horizontal and vertical location of the center of gravity of the tow tractor in air transportable configuration, marked on the silhouette views
3. Shipping weight
4. Loading cubage
5. Overall height, width, and length
6. Front and rear axle loads
7. Tie down information

The information contained on the transportation data plate shall also appear in the Technical Order as it is presented on the plate.

2.27 Identification plate. The tow tractor shall be provided with a permanently-marked identification plate constructed of a non-corroding metal, mounted on the inside of the tow tractor cab on the driver's door panel, and a rear license plate mounting bracket with license plate attaching hardware. The variable information on the identification plate shall be engraved onto the plate by a method that produces a consistent lettering quality. At a minimum, the identification plate shall include the following information:

1. Vehicle Nomenclature
2. Tow Tractor Manufacturer's Model Number
3. Vehicle Identification (serial) Number or VIN
4. Tow Tractor Manufacturer's address and phone number
5. Date of Tow Tractor Manufacture
6. National Stock Number
7. Contract Number
8. Registration Number

The information contained on the identification plate shall also appear in the Technical Order as it is presented on the plate.

2.28 Lubrication data plate. The tow tractor shall be provided with a permanently-marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of

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the tow tractor that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures. The information contained on the lubrication data plate shall also appear in the Technical Order as it is presented on the plate.

2.29 Miscellaneous specifications. The tow tractor shall comply with Society of Automotive Engineers (SAE) specifications ARP1247 and AIR 1316.

2.30 Technical Order (TO) requirements. The contractor shall provide copies of the tow tractor manuals in accordance with the DD Form 1423. The contractor's parts manuals shall list the part numbers for contractor-manufactured parts, plus the part numbers for purchased sub-level components and whole components. The contractor shall also provide with each tow tractor, a product familiarization videotape that verbally and visually provides all information for the operation and routine maintenance of the tow tractor and its components, using the commercial manuals as a baseline. See DD Forms 1423 to this solicitation/contract.

3. REGULATORY REQUIREMENTS.

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practical, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Used, rebuilt, or remanufactured components shall not be incorporated in the tow tractor.

4. PRODUCT CONFORMANCE.

4.1 Product Conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be substantially the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance. The configuration of the offeror's added equipment or features shall be uniform on both the two-wheel drive and four-wheel drive variant unless otherwise approved by the contracting officer. The only differences between the two-wheel drive and four-wheel drive tow tractor variants shall be the chassis differences as delivered by the OE chassis-cab manufacturer. The configuration of the options for an air-system for trailer towing and Air Conditioning (see 6.2 d and 6.2 h) shall be uniform when optioned separately or in combination

4.2 Commercial item requirement. The tow tractor furnished must meet the "commercial item" definition, as specified in FAR 2.101, as of the date of award. The offeror/contractor shall identify all modifications made to their commercial model in order to meet the performance and descriptive requirements of the CID or the referenced documents. In regard to the offered item, the offeror/contractor shall identify any enhancements or improvements to the performance requirements of paragraph 2 of the CID or the referenced documents.

4.3 Verification. Prior to delivery of the first tow tractor of each configuration (two-wheel drive and four-wheel drive), the contractor shall demonstrate or test at his facility, each requirement of the CID or referenced documents.

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4.3.1 Gross vehicle weight (GVW). The first production tow tractor shall be weighed for verification of the minimum GVW.

4.3.2 Overall dimensions. The first production tow tractor shall be measured for verification of the overall dimensions.

4.3.3 Drawbar pull (DBP). The first production tow tractor shall be tested for verification of the DBP on dry level concrete using a load sensing device and a fixed anchoring position. The drawbar position shall be horizontal and level, and the pintle hook shall be adjusted to its lowest height position of 12 inches from the ground.

4.3.4 Power train. The first production tow tractor shall be inspected for verification of the specified power train components. A copy of the OE chassis-cab manufacturer's signed application for approval shall be submitted to the government representative at the first production verification with the vehicle documents for the following components: engine, transmission/transfer case, and axles, to include torque and power ratings, use of Jet A-1 or JP-8 fuels, and emissions compliance. The cooling system shall be certified by the OE chassis-cab manufacturer to satisfy all operating conditions at all ambient temperatures up to the specified, for both the engine and transmission, and any requirements for towing in high ambient temperatures (recommended coolant ratios, oil types and viscosities, etc., for various temperature ranges) shall be submitted to the government representative at the first production verification. The winterization packages shall be certified by the OE chassis-cab manufacturer or the tow tractor manufacturer to satisfy the winterization conditions specified, and a breakdown of the winterization components and requirements for each winterization package (recommended coolant ratios, oil types and viscosities, etc., for the various temperature ranges) shall be submitted to the government representative at the first production verification. Tow tractor component documentation shall include any "special option" codes for the rear axle ratio from the OE chassis-cab manufacturer if applicable.

4.3.5 Driveline and steering. The first production tow tractor shall be demonstrated with a test drive by the government representative. The tow tractor wheelbase shall be measured for verification of the specified dimension. The tow tractor shall be tested for verification of the maximum wall-to-wall turning radius while in four-wheel coordinated steering mode. The fully ballasted first production tow tractor shall be tested as follows to demonstrate compliance with the turning radius specification:

a. Facility. Perform the test on a dry, level, paved area that is free from loose material and is larger in all directions than three times the length of the vehicle being tested.
b. Equipment required.
    1. A device suitable for measuring three times the length of the vehicle being tested with an accuracy of at least ±1-inch.
    2. Markers or marking device suitable for marking the pavement.
    3. A plumb bob suitable for locating a point on the pavement directly below a fixed point on the vehicle.
    4. A test report notebook or similar record forms to be used as a test report work sheet and incorporated into the documentation package.
c. Test Conditions.

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1. The tow tractor shall be fully ballasted and in operational form.
2. The tow tractor steering system shall be fully operational, and the steering linkage stops shall be adjusted to the OE chassis-cab manufacturer's specified production tolerance limits.

d. Test Procedure.
1. The vehicle shall be driven slowly in a full cramp circle (left or right) to establish a steady state in the steering linkage.
2. Continue driving the slow full cramp circle.
3. At approximately three equidistant points (identified as A, B, and C) around the circle, gently stop the vehicle using the service brakes.
4. At each stop, place a plumb bob against the outermost point of the vehicle and mark the spot on the ground directly below where the plumb bob comes to rest.
5. Measure and record the straight line distances between each pair of points, i.e., Lengths AB, BC, and CA.
6. Calculate the wall-to-wall turning radius (R) as follows:

$$R = D / 2 \quad \text{where}$$

$$D = \frac{(AB \times BC \times CA)}{2 \times \sqrt{(S \times (S - AB) \times (S - BC) \times (S - CA))}} \quad \text{and}$$

$$S = (AB + BC + CA) / 2$$

7. Repeat steps (1) through (6) with the vehicle moving in the opposite direction.

4.3.6 Electro-magnetic interference (EMI). The first production tow tractor shall be tested by the contractor for verification of the specified EMI requirements prior to first production verification of the tow tractor. The tow tractor shall be tested for Radiated Emission per RE102 of MIL-STD-461 and Radiated Susceptibility per RS103 of MIL-STD-461. The tow tractor shall be tested for Conducted Susceptibility per CS114, CS115, and CS116 of MIL-STD-461. Test results shall be submitted to the government representative at first production verification.

4.3.7 Mobility. The first production tow tractor shall be tested for verification of the forward and reverse speeds on a flat asphalt or concrete surface without a trailing load, using the tow tractor’s speedometer as the speed-measuring device. The tow tractor shall be tested for verification of its capability to tow a 40,000 pound trailer on a 2.5 % grade at a minimum of 10 MPH on an asphalt or concrete surface, using the tow tractor’s speedometer as the speed measuring device. The tow tractor shall be tested for verification of its capability to operate on 10 percent side slopes and 12 percent grades without fluid spillage.

4.3.8 Suspension. Reserved.

4.3.9 Wheels and tires. The first production tow tractor shall be inspected for verification of the specified wheels and tires. Documentation of the proper wheel and tire selection and deviations

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in tire pressures (if any) with references to the Tire and Rim Association shall be submitted to the government representative at the first production verification.

4.3.10 Braking system. The first production tow tractor shall be inspected for verification of an FMVSS compliant braking system.

4.3.11 Parking brake. The first production tow tractor shall be inspected for verification of a functional, OEM parking brake. If applicable, documentation of any "before" and "after" parking-brake cable relocations, cable-shortening modifications, or associated modification processes shall be presented to the government representative at the first production verification. This documentation shall be verification that the OE chassis-cab manufacturer's design and the tow tractor manufacturer's design modifications are equivalent when producing parking-brake cable tension. If applicable, the documentation should demonstrate equivalence between the original and modified cable mounting locations verses the original and modified cable lengths.

4.3.12 Air system. A schematic of the air system plumbing shall be submitted to the government representative at first production verification.

4.3.13 Electrical system. The first production tow tractor shall be inspected for verification of the specified electrical system.

4.3.14 Chassis modifications. The first production tow tractor shall be inspected for verification of the chassis modifications as specified.

4.3.15 Cab design. The first production tow tractor shall be inspected for verification of the three-man cab, the entrance step and height, cab instrumentation, and specified equipment.

4.3.16 Lighting. The first production tow tractor shall be inspected for verification of the specified lighting and location. Documentation of the double-faced signal lamp's adherence to SAE J588 and the candlepower of the adjustable aim floodlight shall be submitted to the government representative at first production verification.

4.3.17 Pintle hooks. The first production tow tractor shall be inspected for verification of the specified pintle hooks and locations. A strength analysis report shall be submitted to the government representative at first production verification that contains supporting calculations to show that the front and rear pintle hook assemblies and mounting structures are in compliance with their respective strength specifications. The analysis shall include methodology leading to the arrival of values for static and dynamic loading, and ultimately the design load. Additionally, the front-center and rear-center pintle hook assembly and mounting structure calculations shall be supplemented with a Finite Element Analysis (FEA) to further support compliance with the strength specification.

4.3.18 Breakaway cable hookup. The first production tow tractor shall be inspected for verification of the specified breakaway cable hookup and location.

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4.3.19 Safety chain attachment points. The first production tow tractor shall be inspected for verification of the specified rear safety chain attachment points and locations. Documentation of the strength of the safety chain attachment points and mounting (through ratings and/or calculations) shall be submitted to the government representative at first production verification.

4.3.20 Trailer light harness receptacle. The first production tow tractor shall be inspected for verification of the specified J560b electrical receptacle and location. An electrical schematic of the receptacle and its circuitry shall be submitted to the government representative at first production verification.

4.3.21 Fire extinguisher. The first production tow tractor shall be inspected for verification of the specified fire extinguishers and mounting brackets.

4.3.22 Finish. Documentation showing the specifications of the zinc-rich primer used in the tow tractor's production shall be submitted to the government representative at the first production verification. Color sample cards with the specified colors shall be submitted at the first production verification. Tow tractor finish documentation shall include any "special option" codes for the top coat finish from the OE chassis-cab manufacturer if applicable.

4.3.23 Corrosion protection. The first production tow tractor shall be inspected for verification of the specified corrosion protection. Documentation of the application process and certification of the level of corrosion treatment shall be submitted to the government representative at the first production verification.

4.3.24 Visibility. The first production tow tractor shall be inspected for verification of the specified visibility. Measurements of the defined visibility plane shall be made, and verification of the visibility shall be performed

4.3.25 Air transportability. An air transportability report shall be submitted to the government representative at first production verification to begin the air transportability certification process. The report shall outline the steps necessary to load the tow tractor onto the specified aircraft, including diagrams, drawings, or instructions of the tow tractor preparation, loading, shoring, and tie down procedures (including engineering analysis of the tie down devices) for both the tow tractor and any removable ballasts, if applicable, along with critical dimensions, tow tractor weights, axle weights, etc.

4.3.26 Transportation data plate. The first production tow tractor shall be inspected for verification of the specified transportation data plate.

4.3.27 Identification plate. The first production tow tractor shall be inspected for verification of the specified identification plate and a rear license plate mounting bracket.

4.3.28 Lubrication data plate. The first production tow tractor shall be inspected for verification of the specified lubrication data plate.

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4.3.29 Miscellaneous specifications. The first production tow tractor shall be inspected for verification of compliance with SAE specifications ARP1247 and AIR 1363.

4.4 Reconfiguration criteria. Failure to comply with the requirements of this CID, reference documents, or the tests shall be cause for reconfiguration and re-demonstration. Causes for reconfiguration shall include: structural failure; non-correctable misalignments; component interference; conditions presenting a safety hazard to the user or maintainer: instability during operation, spillage of fuel or coolant, overheating, and driveline vibration or shudder due to sub-optimal driveline angles.

5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order (see 6.2 b).

6. NOTES.

6.1 Sources of documents.

6.1.1 Department of Defense and Federal documents may be obtained from the Document Automation and Production Service, 700 Robbins Avenue, Philadelphia PA 19111-5094, or online at http://assist.daps.dla.mil .

6.1.2 FAR. The Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at http://www.arnet.gov/far.

6.1.3 SAE documents may be obtained from SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096, or online at http://www.sae.org.

6.2 Ordering data. The purchaser must specify the following at the time of purchase:

a. Title, number, and revision letter, and date of this document.
b. Packaging requirements (see 5.1).
c. When heavy-duty (type A) winterization is required (see 2.4).
d. When an air-system for trailer towing is required (see 2.12 and 2.15).
e. Color required (see 2.22).
f. When four-wheel drive is required (see 2.5).
g. When driver’s side spotlight is required (see 2.16).
h. When Air Conditioning is required (see 2.15).

6.3 Subject term (key word) listing.

Aircraft parking
Bobtail
FLTT
Flight line
Ground handling
Tug

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MILITARY INTERESTS

Custodians:
Air Force – 99
Army – AV

Preparing activity:
Air Force – 84

Agent
Air Force – 99

(Project 1740-2006-004)

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